

For Week 3, include concepts such as linear regression with forward and backward selection, PCR, and PLSR. Complete your Jupyter Notebook homework by 11:59 pm ET on Sunday.

```
In [1]: import pandas as pd
from imblearn.under_sampling import RandomUnderSampler
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import classification_report
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import make_pipeline
from sklearn.decomposition import PCA
from sklearn.cross_decomposition import PLSRegression
from sklearn.metrics import average_precision_score
```

```
In [2]: random_state=42
```

## PPP

### Data

```
In [3]: df = pd.read_csv('C:/Users/caleb/Projects/BU Spring 2026/Module-B-semester-2/Milest
df
```

```
Out[3]:
```

|        | ProcessingMethod | BorrowerState | LoanStatus | Term | InitialApprovalAmount | Curru |
|--------|------------------|---------------|------------|------|-----------------------|-------|
| 0      | 0.0              | 48.0          | 2.0        | 24   | 769358.78             |       |
| 1      | 0.0              | 48.0          | 2.0        | 24   | 736927.79             |       |
| 2      | 0.0              | 48.0          | 2.0        | 24   | 691355.00             |       |
| 3      | 0.0              | 48.0          | 2.0        | 24   | 499871.00             |       |
| 4      | 0.0              | 48.0          | 2.0        | 24   | 367437.00             |       |
| ...    | ...              | ...           | ...        | ...  | ...                   |       |
| 968527 | 0.0              | 56.0          | 2.0        | 24   | 150000.00             |       |
| 968528 | 0.0              | 56.0          | 2.0        | 24   | 150000.00             |       |
| 968529 | 1.0              | 56.0          | 2.0        | 60   | 150000.00             |       |
| 968530 | 0.0              | 56.0          | 2.0        | 60   | 150000.00             |       |
| 968531 | 0.0              | 56.0          | 2.0        | 24   | 150000.00             |       |

968532 rows × 42 columns



## Undersampling

```
In [4]: from collections import Counter
```

```
X = df.drop(columns='Fraud')
y = df['Fraud']
print("Original dataset shape:", Counter(y))
rus = RandomUnderSampler(sampling_strategy=0.1, random_state=random_state)
X_resampled, y_resampled = rus.fit_resample(X, y)

print("Resampled dataset shape:", Counter(y_resampled))
```

Original dataset shape: Counter({0: 968437, 1: 95})

Resampled dataset shape: Counter({0: 950, 1: 95})

## Split Data

```
In [5]: X_train, X_test, y_train, y_test = train_test_split(X_resampled,
                                                            y_resampled,
                                                            test_size=0.2,
                                                            stratify=y_resampled, # Maintain
                                                            random_state=random_state
                                                            )
```

## PCR

```
In [17]: pcr_model = make_pipeline(
            StandardScaler(),
            PCA(),
            LogisticRegression()
        )

pcr_model.fit(X_train, y_train)
y_pred = pcr_model.predict(X_test)
```

```
In [18]: print(classification_report(y_true=y_test, y_pred=y_pred))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.97      | 0.97   | 0.97     | 190     |
| 1            | 0.72      | 0.68   | 0.70     | 19      |
| accuracy     |           |        | 0.95     | 209     |
| macro avg    | 0.85      | 0.83   | 0.84     | 209     |
| weighted avg | 0.95      | 0.95   | 0.95     | 209     |

```
In [19]: average_precision_score(y_true=y_test, y_score=y_pred)
```

```
Out[19]: 0.5228601807549176
```

## PLSR

```
In [20]: pls = PLSRegression(n_components=2)
pls.fit(X_train, y_train)

y_pred_continuous = pls.predict(X_test).ravel()
y_pred_classes = (y_pred_continuous > 0.5).astype(int) # PLSR is regression, we need to threshold
print(classification_report(y_test, y_pred_classes))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.97      | 0.97   | 0.97     | 190     |
| 1            | 0.72      | 0.68   | 0.70     | 19      |
| accuracy     |           |        | 0.95     | 209     |
| macro avg    | 0.85      | 0.83   | 0.84     | 209     |
| weighted avg | 0.95      | 0.95   | 0.95     | 209     |

```
In [21]: average_precision_score(y_true=y_test, y_score=y_pred_classes)
```

```
Out[21]: 0.5228601807549176
```

## IBM

### Data

```
In [22]: df = pd.read_csv('ibm_hi_small_trans_cleaned_semester3.csv')
df
```

Out[22]:

|         | Is<br>Laundering | Amount_Received_USD | Amount_Paid_USD | Receiving<br>Currency_Bitcoin | Re<br>Currenc |
|---------|------------------|---------------------|-----------------|-------------------------------|---------------|
| 0       | 0                | 3697.340000         | 3697.340000     | 0.0                           |               |
| 1       | 0                | 0.010000            | 0.010000        | 0.0                           |               |
| 2       | 0                | 14675.570000        | 14675.570000    | 0.0                           |               |
| 3       | 0                | 2806.970000         | 2806.970000     | 0.0                           |               |
| 4       | 0                | 36682.970000        | 36682.970000    | 0.0                           |               |
| ...     | ...              | ...                 | ...             | ...                           |               |
| 5078340 | 0                | 3107.386389         | 3107.386389     | 1.0                           |               |
| 5078341 | 0                | 2168.020464         | 2168.020464     | 1.0                           |               |
| 5078342 | 0                | 100.011894          | 100.011894      | 1.0                           |               |
| 5078343 | 0                | 770.280058          | 770.280058      | 1.0                           |               |
| 5078344 | 0                | 5653.900141         | 5653.900141     | 1.0                           |               |

5078345 rows × 39 columns



## Undersampling

In [23]: `from collections import Counter`

```
X = df.drop(columns='Is Laundering')
y = df['Is Laundering']
print("Original dataset shape:", Counter(y))
rus = RandomUnderSampler(sampling_strategy=0.1, random_state=random_state)
X_resampled, y_resampled = rus.fit_resample(X, y)

print("Resampled dataset shape:", Counter(y_resampled))
```

Original dataset shape: Counter({0: 5073168, 1: 5177})

Resampled dataset shape: Counter({0: 51770, 1: 5177})

## Split Data

```
In [24]: X_train, X_test, y_train, y_test = train_test_split(X_resampled,
                                                             y_resampled,
                                                             test_size=0.2,
                                                             stratify=y_resampled, # Maintain
                                                             random_state=random_state
                                                             )
```

## PCR

```
In [25]: pcr_model = make_pipeline(
          StandardScaler(),
          PCA(),
          LogisticRegression()
        )

pcr_model.fit(X_train, y_train)
y_pred = pcr_model.predict(X_test)
```

```
In [26]: print(classification_report(y_true=y_test, y_pred=y_pred))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.92      | 1.00   | 0.95     | 10355   |
| 1            | 0.80      | 0.08   | 0.14     | 1035    |
| accuracy     |           |        | 0.91     | 11390   |
| macro avg    | 0.86      | 0.54   | 0.55     | 11390   |
| weighted avg | 0.90      | 0.91   | 0.88     | 11390   |

```
In [27]: average_precision_score(y_true=y_test, y_score=y_pred)
```

```
Out[27]: 0.1440033248860868
```

## PLSR

```
In [34]: pls = PLSRegression(n_components=2)
pls.fit(X_train, y_train)

y_pred_continuous = pls.predict(X_test).ravel()
y_pred_classes = (y_pred_continuous > 0.5).astype(int) # PLSR is regression, we need to
print(classification_report(y_test, y_pred_classes))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.92      | 1.00   | 0.95     | 10355   |
| 1            | 0.79      | 0.07   | 0.14     | 1035    |
| accuracy     |           |        | 0.91     | 11390   |
| macro avg    | 0.85      | 0.54   | 0.55     | 11390   |
| weighted avg | 0.90      | 0.91   | 0.88     | 11390   |

```
In [35]: average_precision_score(y_true=y_test, y_score=y_pred_classes)
```

```
Out[35]: 0.1431655933601344
```

## Credit Card

### Data

```
In [36]: df = pd.read_csv('C:/Users/caleb/Projects/BU Spring 2026/Module-B-semester-2/Milest
df
```

```
Out[36]:
```

|               | Time     | V1         | V2        | V3        | V4        | V5        | V6        |
|---------------|----------|------------|-----------|-----------|-----------|-----------|-----------|
| <b>0</b>      | 0.0      | -1.359807  | -0.072781 | 2.536347  | 1.378155  | -0.338321 | 0.462388  |
| <b>1</b>      | 0.0      | 1.191857   | 0.266151  | 0.166480  | 0.448154  | 0.060018  | -0.082361 |
| <b>2</b>      | 1.0      | -1.358354  | -1.340163 | 1.773209  | 0.379780  | -0.503198 | 1.800499  |
| <b>3</b>      | 1.0      | -0.966272  | -0.185226 | 1.792993  | -0.863291 | -0.010309 | 1.247203  |
| <b>4</b>      | 2.0      | -1.158233  | 0.877737  | 1.548718  | 0.403034  | -0.407193 | 0.095921  |
| ...           | ...      | ...        | ...       | ...       | ...       | ...       | ...       |
| <b>284802</b> | 172786.0 | -11.881118 | 10.071785 | -9.834783 | -2.066656 | -5.364473 | -2.606837 |
| <b>284803</b> | 172787.0 | -0.732789  | -0.055080 | 2.035030  | -0.738589 | 0.868229  | 1.058415  |
| <b>284804</b> | 172788.0 | 1.919565   | -0.301254 | -3.249640 | -0.557828 | 2.630515  | 3.031260  |
| <b>284805</b> | 172788.0 | -0.240440  | 0.530483  | 0.702510  | 0.689799  | -0.377961 | 0.623708  |
| <b>284806</b> | 172792.0 | -0.533413  | -0.189733 | 0.703337  | -0.506271 | -0.012546 | -0.649617 |

284807 rows × 31 columns

## Undersampling

```
In [37]: from collections import Counter

X = df.drop(columns='Class')
y = df['Class']
print("Original dataset shape:", Counter(y))
rus = RandomUnderSampler(sampling_strategy=0.1, random_state=random_state)
X_resampled, y_resampled = rus.fit_resample(X, y)

print("Resampled dataset shape:", Counter(y_resampled))
```

```
Original dataset shape: Counter({0: 284315, 1: 492})
Resampled dataset shape: Counter({0: 4920, 1: 492})
```

## Split Data

[illegible]

## PCR

```
In [39]: pcr_model = make_pipeline(  
        StandardScaler(),  
        PCA(),  
        LogisticRegression()  
    )  
  
    pcr_model.fit(X_train, y_train)  
    y_pred = pcr_model.predict(X_test)
```

```
In [40]: print(classification_report(y_true=y_test, y_pred=y_pred))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.99      | 1.00   | 0.99     | 985     |
| 1            | 0.96      | 0.89   | 0.92     | 98      |
| accuracy     |           |        | 0.99     | 1083    |
| macro avg    | 0.97      | 0.94   | 0.96     | 1083    |
| weighted avg | 0.99      | 0.99   | 0.99     | 1083    |

```
In [43]: average_precision_score(y_true=y_test, y_score=y_pred)
```

```
Out[43]: 0.8588898711291159
```

## PLSR

```
In [47]: pls = PLSRegression(n_components=2)  
    pls.fit(X_train, y_train)  
  
    y_pred_continuous = pls.predict(X_test).ravel()  
    y_pred_classes = (y_pred_continuous > 0.5).astype(int) # PLSR is regression, we need to threshold  
  
    print(classification_report(y_test, y_pred_classes))
```

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| 0            | 0.96      | 1.00   | 0.98     | 985     |
| 1            | 0.97      | 0.60   | 0.74     | 98      |
| accuracy     |           |        | 0.96     | 1083    |
| macro avg    | 0.96      | 0.80   | 0.86     | 1083    |
| weighted avg | 0.96      | 0.96   | 0.96     | 1083    |

```
In [48]: average_precision_score(y_true=y_test, y_score=y_pred_classes)
```

```
Out[48]: 0.6183128535006936
```